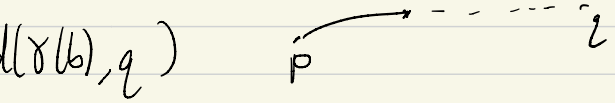
Thm (Hopf-Rinow 1931)
For a connected Riem mfld, metric and geodesic completeness are equivalent.

metric $\Rightarrow$ geodesic ✓
Need to show geodesic $\Rightarrow$ metric

Lemma Suppose $\exp_p$ is defined on all of T_pM (geodesic rays from p exist for $t \to \infty$).
Then for all $q \in M$, there exists a length-minimizing geodesic p to q

Proof: We say a geodesic segment $\gamma : [0,b] \to M$ with $\gamma(0) = p$ aims at q if

$$d(p,q) = d(p,\gamma(b)) + d(\gamma(b),q)$$



Claim 1: such segments exist

Choose $\varepsilon > 0$ so that the closed geodesic ball $\overline{B_\varepsilon(p)}$ is well-defined, set $S_\varepsilon = \partial B_\varepsilon(p)$

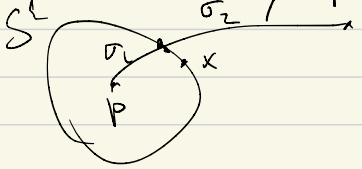
If $q \in \overline{B_\varepsilon(p)}$ then we already know that the radial geodesic is minimizing by the Gauss Lemma

Assume $q \notin \overline{B_\varepsilon(p)}$

S_ε is compact because $\exp_p$ is a homeomorphism $\{|v| = \varepsilon\} \to \overline{B_\varepsilon(p)}$

$\Rightarrow$ the distance $d(\cdot, q)$ is minimized over S_ε at some pt. $x \in S_\varepsilon$

Consider any pcwise regular path $\sigma : p$ to q



$\ell(\sigma_1) \geq \varepsilon$ since endpt lies on S_ε

$\ell(\sigma_2) \geq d(x,q)$ by choice of x

$\rightarrow d(p,q) \geq \varepsilon + d(x,q)$

Since $d(p,q) \leq \varepsilon + d(x,q)$ by triangle,

$d(p,q) = \varepsilon + d(x,q)$

$\therefore \gamma$ aims at q

$=$

Now set $T = d(p,q)$ and let

$I = \{ l(\gamma) : \gamma \text{ is a geodesic aimed at } q \} \subset (0,T]$

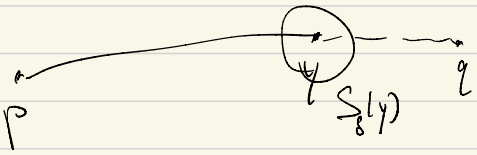
Claim 1 $\Rightarrow I \neq \emptyset$

continuity of distance $\rightarrow I = [0,b]$ for some $b \leq T$

Claim 2: $I = (0,T]$

Given $t<T$ in I let γ be the geodesic and $\gamma(t) = y$

Choose δ so that $\overline{B_\delta(y)}$ is a geodesic ball, $q \notin \overline{B_\delta(y)}$



By claim 1, we can choose $z \in S_\delta(y)$ so that the radial geodesic τ aims at q from y

Now we have $T = t + d(y,q)$ (γ aims at q)

and $d(y,q) = \delta + d(z,q)$ (τ aims at q)

$\therefore d(z,q) = T - t - \delta$

By triangle $\underbrace{d(p,q)}_{T} \leq d(p,z) + \underbrace{d(z,q)}_{T-t-\delta}$

$\rightarrow d(p,z) \geq t + \delta$

Since $\gamma + \tau$ has this length, $\gamma + \tau$ is a length minimizing curve p to z

Variation thm $\Rightarrow$ $\gamma_{t\tau}$ is regular geodesic

in $t \in S \subseteq I$,

This proves $I = [0, T]$

$=$

Thm (improved part 2 of Hopf-Rinow)

Suppose $\exp_p$ exists on all of T_pM for some $p \in M$,

Then M is metrically complete.

Proof: Suppose $\{q_i\} \subset M$ is a Cauchy sequence

By the Lemma, each $q_i = \exp_p(V_i)$ where

$$\|V_i\| = d(p, q_i)$$

$\{q_i\}$ Cauchy $\Rightarrow$ bound $\|V_i\| \leq C$ for all i

$\Rightarrow \exists$ a subsequence V_{i_k} so that

$$V_{i_k} \xrightarrow{k \to \infty} V \text{ in } \{\|V\| \leq C\}$$

$\rightarrow$ $q_{i_k} \longrightarrow q := \exp_p(V)$ in M

$\therefore$ all of $\{q_i\}$ converges to q.

$=$

From the proof of Hopf-Rinow, we can see another equivalent def

Cor: M is complete if and only if any two points can be joined by a length-minimizing geodesic.

Curvature

The Riemannian curvature endomorphism is a map

$$TM \times TM \times TM \to TM$$

given by $R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$

example: Euclidean $\mathbb{R}^n$, Christoffel symbols $= 0$

$\Rightarrow$ for $Z = v^j \partial_j$,

$$\nabla_Y Z = (Yv^j)\partial_j \quad \text{(no connection)}$$

$$\Rightarrow R(X,Y)Z = X(Yv^j)\partial_j - Y(Xv^j)\partial_j - ([X,Y]v^j)\partial_j$$

$$= 0$$

Prop. R is a $\binom{3}{1}$ tensor which is invariant under local isometries

Proof: tensor means multilinear over $C^\infty(M)$

i.e. (i) $R(X_1 + X_2, Y)Z = R(X_1,Y)Z + R(X_2,Y)Z$

and (ii) $R(fX,Y)Z = f\,R(X,Y)Z$ for $f \in C^\infty(M)$

(similarly for the Y and Z variables)

These follow from the properties of the connection

example: $R(fX,Y)Z = \nabla_{fX}\nabla_Y Z - \nabla_Y \nabla_{fX} Z - \nabla_{[fX,Y]} Z$

note $[fX,Y] = fXY - Y(fX)$

$= f[X,Y] - (Yf)X \qquad Yf \in C^\infty(M)$

$$R(fX,Y)Z = f\nabla_X \nabla_Y Z - \nabla_Y(f\nabla_X Z) - \nabla_{f[X,Y]} Z + \nabla_{(Yf)X} Z$$

$$= fR(X,Y)Z - (Yf)\nabla_X Z + (Yf)\nabla_X Z$$

(similar for other cases)

Invariance is a long but straightforward calculation